

Jose Salgado

Senior AI/ML Engineer

jose.eng910@gmail.com | (718) 223-4174 | [Github](#) | [LinkedIn](#)

Summary

Senior AI/ML Engineer with over 10 years of experience delivering state-of-the-art AI solutions in GenAI, Large Language Models (LLMs), Reinforcement Learning (RL), and enterprise-level applications. Expertise in AI Systems Architecture, MLOps, and AI deployment, delivering scalable products in sectors such as healthcare, finance, and logistics. Skilled in transforming cutting-edge research into secure, scalable, and efficient AI systems, leveraging Claude, GPT, LangChain, LangGraph, Weaviate, Vertex AI, AWS, and Google Cloud Platform (GCP). Proven track record in leading and executing Generative AI, Retrieval-Augmented Generation (RAG), and multi-agent systems for scalable enterprise solutions.

SKILLS

- **LLM / GenAI:** Claude 1/2/3, GPT-3, Codex, LLM inference, high-concurrency APIs, streaming (SSE/WebSockets), tool/function calling, JSON schema outputs, prompt engineering, prompt chaining, context windows, token budgeting, caching, guardrails, jailbreak resistance
- **RAG / Retrieval:** RAG, embeddings, chunking, parsing, metadata filtering, hybrid search (BM25 + dense), reranking, query rewriting, multi-hop retrieval, context packing, citation grounding, hallucination mitigation, faithfulness evaluation, Pinecone, Weaviate, vector indexing, namespaces, tenant isolation, ACL-aware retrieval, incremental reindexing, upserts, TTL, embedding refresh
- **Agents / Orchestration:** Multi-agent systems, planner/critic/executor, tool routing, durable state, retries/fallbacks, HITL approvals, policy gates, audit trails, LangChain, LangGraph, LangSmith, tracing, eval harnesses, golden sets, regression suites, adversarial test suites, continuous evaluation
- **MLOps / Infra:** CI/CD, model/prompt/config versioning, artifact registries, reproducible builds, environment pinning, canary deploys, rollbacks, feature flags, AWS, Kubernetes, Terraform, containers, autoscaling, GPU/TPU scheduling, capacity planning, cost governance, secrets management, IAM/RBAC
- **Training / Optimization:** TensorFlow, PyTorch, fine-tuning, domain adaptation, dataset curation, deduplication, checkpointing, mixed precision, profiling, memory optimization, pruning, quantization, batching, KV-cache management, prefill/decode optimization
- **RL / Safety:** Self-play, reward shaping, reward modeling, RLHF concepts, red teaming, prompt injection defense, MCTS, DQN, distributional RL, evaluation at scale, robustness testing
- **Data / Platforms:** Databricks, Apache Spark, Delta Lake, streaming pipelines, ETL/ELT, backfills, data quality checks, SLAs, MLflow, experiment tracking, model registry, reproducibility

PROFESSIONAL EXPERIENCE

Senior AI/ML Engineer, *Anthropic*

2022 – 2025

Role: Lead Engineer for Generative AI Deployment, Safety, and MLOps

- Claude GenAI Product Leadership & Enterprise Application

- Led enterprise rollouts of Claude across regulated customers, focusing on security basics that actually unblock adoption: tenant isolation, RBAC, audit logs, private connectivity patterns, and safe data handling.
- Built a shared “LLM gateway” layer for teams: request routing, quotas, streaming, prompt/version management, safety filters, and structured output validation - so new Claude features shipped fast and consistently.
- Delivered production-ready patterns for high-stakes workflows: citation-first answers, safe tool execution, prompt-injection mitigation, and human review for sensitive actions like improving reliability and reducing escalations.

- RAG (Retrieval-Augmented Generation) Systems Architecture

- Designed a production RAG stack using Claude + Weaviate/Pinecone: ingestion, chunking, embeddings, metadata/ACL propagation, retrieval + reranking, context assembly, and grounded citations.
- Built evaluation around retrieval + grounding quality: golden sets, regression tests, and tracking for retrieval hit rate and answer faithfulness—keeping quality stable as corpora changed.
- Shipped enterprise chat experiences with scoped retrieval (user/team permissions), incremental indexing, stale-content handling, and fallbacks (ask-clarify/escalate/refuse) for safer outcomes.

- MLOps and Scalable GenAI Solutions

- Architected CI/CD for GenAI services on AWS/Kubernetes/Terraform: environment promotion, canary rollouts, automated rollback triggers, and config-driven routing for model/prompt updates.
- Implemented monitoring that helped in production: token/cost dashboards, p95/p99 latency, tool-call tracing, retrieval metrics, and safety event logs using Prometheus/Grafana/OpenTelemetry.
- Owned performance + cost tuning for inference: dynamic batching, caching, warm pools, and capacity planning to keep latency predictable under enterprise traffic.

- Multi-Agent Systems Development & Automation

- Used LangChain + LangGraph to ship MVP agent workflows quickly, then hardened them with durable state, retries/timeouts, and strict tool contracts for production reliability.
- Built secure tool integrations for agents (internal APIs, ticketing, knowledge systems) using schema validation, permission gates, and audit trails.
- Standardized agent evaluation in LangSmith with scenario suites, adversarial prompts, and tool-failure simulations tied to release gates—cutting manual QA and preventing regressions.

Machine Learning Engineer, OpenAI

2019 – 2022

Role: Lead Engineer for LLM Training, RL Integration, and API Deployment

- GPT-3 & Codex Infrastructure Development

- Built and owned core pieces of the GPT-3 serving stack: request routing, streaming generation, rate limits/quotas, reliability hardening, and production runbooks to keep the API stable under fast-growing traffic.

- Worked on scaling mechanics that mattered in practice: load balancing, replica autoscaling, safe rollouts/rollbacks, per-tenant isolation, and “noisy neighbor” controls for large customers.

- Improved Codex quality end-to-end: data curation + filtering, fine-tuning runs, evaluation sets for code tasks (pass@k-style), and IDE-facing latency fixes that made completions feel responsive in editor workflows.

- Reinforcement Learning (RL) Systems and Safety Enhancements

- Contributed to large-scale RL training infrastructure: distributed rollout collection, replay/trajectory storage, reward shaping pipelines, and evaluation harnesses designed to catch regressions early.

- Built debugging and measurement tools around RL runs: episode analytics, reward decomposition, ablation toggles, and failure mode tagging - so training issues were diagnosable instead of “it diverged again.”

- Implemented safety checks for RL agents in simulated environments: constraint tracking, adversarial scenario generation, and automated “do-not-do” regression tests before promoting new policies.

- Reinforcement Learning (RL) Systems and Safety Enhancements

- Applied RL methods to language-model alignment work: preference data handling, reward-model training loops, policy optimization runs, and evaluation suites focused on helpfulness/harmlessness tradeoffs.

- Developed adversarial test sets for LLM behavior (prompt injection, policy evasion, unsafe instructions) and wired them into release gates so regressions blocked deployments automatically.

- Built “behavior monitoring” for production models: trend detection on unsafe outputs, sampling-based audits, and rapid rollback playbooks when metrics moved in the wrong direction.

- AI Model Optimization & Deployment

- Drove training and inference efficiency work: memory profiling, mixed precision, gradient checkpointing, throughput tuning, and practical model-optimization tradeoffs that cut iteration time for researchers.

- Optimized inference latency for real users: batching, caching, tail-latency reduction (p95/p99), and stability fixes around long prompts, timeouts, and partial-stream failures.

- Took research prototypes to production: packaging + versioning, reproducible configs, offline eval, staging, production promotion, and integration into the public API with monitoring and rollback safety.

AI Engineer, Databricks

2017 – 2019

Role: AI Engineer | Customer-Facing AI Solutions & Machine Learning Infrastructure

- AI Model Development & Custom Client Solutions

- Built and delivered end-to-end ML solutions on the Databricks platform for enterprise customers, working directly with client data teams to translate business problems into deployable models.

- Developed supervised learning models for use cases such as predictive maintenance, churn prediction, and risk scoring using Spark ML and custom Python pipelines, with models designed to scale across large datasets.

- Worked hands-on with customers to productionize models, including feature engineering, model validation, batch scoring jobs, and integration with downstream business systems.

- Optimizing Data Pipelines and Real-Time Analytics

- Designed and optimized large-scale Spark pipelines for real-time and near-real-time analytics, supporting fraud detection, demand forecasting, and operational reporting use cases.
- Implemented structured streaming pipelines with Delta Lake to support continuously updating datasets, handling late-arriving data, schema evolution, and data quality checks.
- Improved pipeline performance and reliability through partitioning strategies, caching, job tuning, and monitoring, enabling faster iteration and more timely business insights.

- MLOps Implementation with MLflow

- Implemented MLflow-based workflows for experiment tracking, model versioning, and reproducible training runs, helping teams move from ad hoc notebooks to repeatable ML processes.
- Set up model packaging and deployment patterns for batch inference and scheduled retraining jobs, reducing manual deployment effort and operational risk.
- Worked with customer teams to establish basic ML governance practices, including experiment comparison, artifact management, and promotion of models from development to production environments.

AI Researcher, Google DeepMind

2015 – 2017

Role: AI Researcher Specializing in RL, Deep Learning, and NLP

- AlphaGo Contributions

- Worked on policy and value network training pipelines for AlphaGo, focusing on stability, sample efficiency, and integration with Monte Carlo Tree Search (MCTS) during large-scale self-play training.
- Contributed to improvements in MCTS components, including rollout evaluation, policy priors, and value backpropagation, supporting stronger move selection under limited compute budgets.
- Ran large-scale experiments on Atari and control benchmarks using DQN and Distributional RL variants, analyzing generalization failures and improving robustness on high-dimensional state spaces.

- WaveNet & AlphaFold

- Contributed to early WaveNet research by experimenting with architectural variants, training stability techniques, and data preprocessing pipelines for neural text-to-speech models.
- Improved audio quality through loss-function tuning, conditioning strategies, and evaluation workflows, helping push neural TTS beyond traditional concatenative and parametric methods.
- Worked on early AlphaFold I modeling efforts, supporting data preparation, feature extraction, and model evaluation for protein structure prediction, collaborating closely with biologists to validate results against known structures.

EDUCATION

Rice University, Bachelor's Degree

2011 – 2015 | Houston, TX